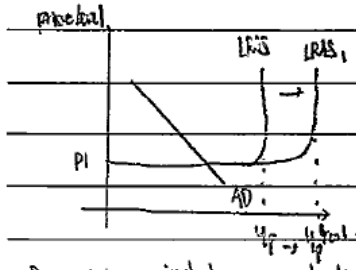
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a) Supply-side policies are government policies that are intended to influence the aggregate supply of an economy, which specifically consists of interventionist and market-based policies. Supply side policies can be used to not only improve domestic competition through low cost of production, but also international competitiveness ~~defined as~~ which consists price and quality competitiveness.



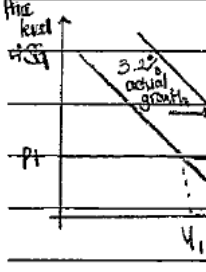
First, supply side policies can encourage greater domestic competition through low cost of production. Market-based policies, which consist of deregulation and privatisation, ~~can be~~ can be useful in achieving this objective.

Deregulation includes reduction or abolition of the ~~minimum wage~~ direct tax on firm's production of goods and services. As firms have a lower direct tax, they will be able to ~~generate~~ ~~more~~ further invest on the R/D of the production. This can be on physical, social, or human capital. As a result, the productivities of workers or machines, defined as worker per output per period of time, will significantly increase, and firms can produce goods and services in a lower cost per goods or services produced than before. This will increase the potential output of an economy, which is illustrated in the diagram as an increase of real GDP from Yp to Yp'. Hence, low cost of production leads to greater competition in the market, as consumers will incentivised to purchase goods and services with a lower price.

Second, supply side policies can encourage international competitiveness of a country. International competitiveness consists of price and quality competitiveness, which is best achieved by the interventionist policies in the economy. Interventionist policy includes government spending on merit goods, such as healthcare and education. When labour receives these ~~governm~~ benefits in a long run, they will develop adequate or sufficient skills to be employed and escape structural unemployment.

These highly skilled workers will then be capable of generating a larger potential output (Y_P') due to their high productivity. With this, the firms can produce goods and services in a lower cost of production, sometimes lower or equal to the world price. The firms therefore then possesses price competitiveness, since consumers from foreign countries will be incentivised to purchase cheap imported goods and services. ~~Also, a high level of education and healthcare can attract foreign nationals to live in this country, such as Singapore with extensive 3.5 trillion healthcare and educating infrastructure spending announced in 2018 over the next 10 years.~~ This In a long run, with this advantage in the international market, the firms can continuously invest on improving the quality of their product through R/D. This will allow firms to be equipped with quality competitiveness, which will further stimulate higher revenue gains for firms and higher income for workers.

b) Interventionist supply side policies are government policies that are intended to influence the aggregate supply of an economy through government spending on ~~free~~ merit goods such as healthcare and education. This policy may be effective in reducing various types of unemployment, such as structural, seasonal, or cyclical unemployment, but has drawbacks that may diminish the effect of the policy. Unemployment refers to the people who doesn't have a job but are willing to and able to work in the labour force.



In 2018, Singapore government has announced a 3.5 trillion dollar investment on healthcare and education infrastructure over 10years. This policy is intended to train workers with novel technologies such as the 5G-AI system ~~in the SG economy. 4th industrial revolution~~ This has caused ~~a~~ 6% reduction in unemployment rate an increase in the long-run aggregate supply (LRAS) of an economy, ~~resulting~~ illustrated as an outward shift of the LRAS curve from LRAS to LRAS1. This has resulted in an increase of the potential output of the economy, illustrated as an increase from Y_P to Y_{P1}. This indicates that workers are receiving adequate healthcare and education to build skills that are required in the labour-force ~~or~~ and improving productivity, defined as worker per output per period of time. ~~As this more~~ This helps to reduce the natural rate of unemployment, specifically structural unemployment, defined as an unemployment due to a mismatch of skills. With training and education, those who lacked skills and experienced hysteresis in the long run could have returned to workforce and receive income. Also, due to improved productivity of labour, the firm gains a larger revenue. It can then invest back on revenue

developing its R/D and quality of the goods and services. Investment, being a component of aggregate demand (AD), leads to an increase in AD, which is illustrated as a outward shift of the AD curve from AD to AD$_1$. As a result, the real GDP increases, from Y_1 to Y_2. ~~to the ear~~ This is effective in reducing cyclical unemployment, as in case of Singapore there were a reduction of 4.6% of unemployment rate from 2018 to ~~2019~~ 2021 Q3. However, the major flaw of the interventionist supply side policies is that it involves a budget deficit. This budget deficit may harm the future generation's ability to meet their macro economic objectives, including a failure to provide pensions or unemployment benefits. This will entrap low income people in the poverty trap, which will eventually create more unemployment ~~due to a lack of growth in firms~~ Especially, those low income households will not receive education (can't afford education without pensions), which will generate generational poverty along with hysteresis. This will worsen the (structural) unemployment of an economy.

Another policy that can be suggested to ~~resolve or~~ reduce the rate of unemployment is monetary policy, especially interest rates. Interest rates are either cost of borrowings or reward of savings. If there ~~is~~ are low interest rates, consumers are encouraged to ~~borro~~ save less in the banking sector but also businesses are encouraged to make borrowings from the bank. Hence, there will be more consumption and investment on capital equipments in an economy. ~~As~~ Both consumption and investment being a component of AD, the AD will increase as a result, and the real GDP will increase. This increase will be stimulated by the accelerator effect, in which more investment allows cheaper ~~cos~~ price which then promotes more consumption, and the local multiplier effect. In a neo-classical model, the increase in AD will therefore alleviate cyclical unemployment. Also, ~~by closing the~~ assuming the economy already has a deflationary gap, the economy can close this gap through an increase in AD. This will also reduce natural rate of unemployment, which consists of seasonal, structural, and frictional unemployment. Knowing these benefits, the FED (Federal Reserve) has lowered its interest rates to 0-25% in 2020 Q2 when the global pandemic arrived. This measure was intended to stimulate the economy out of recession and reduce unemployment. It indeed succeeded, when the unemployment rate indicated 2.3% in Q3 2021. However, there are also drawbacks for the alternative policy - monetary policy. In a neo-classical model, the utilisation of expansionary monetary policy is followed by a demand-pull inflation. This ~~will~~ could possibly harm both consumers and producers, as ~~they~~ the price of goods and services start to increase.

This has been pointed out as a serious issue in 2022, as FED realised there was a mixture of cost-push inflation with demand-pull inflation driven by not only the Ukrainian war and the pandemic but also the expansionary monetary policies. This inflation has increased the CPI (Consumer price index) by 5.8%, which forced FED to incrementally increase interest rates. In May 5th 2022, FED announced that they will increase interest rates up to 1.25% in order to mitigate high inflation. This can be detrimental, as this indicates the cost of borrowing will increase for businesses. They therefore have a risk of going bankrupt by not being able to pay debts, and workers will then be unemployed again.

In conclusion, the statement that supply side interventionist policies is the most effective way of reducing rate of unemployment is debatable. In a long run, it is evident that an increase in workers' productivity will increase LRAS along with AD, which will alleviate both cyclical and natural rate of unemployment. However, the budget deficit followed by the low unemployment can diminish the effects of supply-side policies in reducing unemployment. Especially in the era of pandemic, those budget deficits are affordable by few countries. Singapore, known for its 31% direct progressive income tax for the highest braket, has collected enough government budget before the pandemic, which allowed them to continue its implementation of supply side policies. However, a country next to Singapore - Malaysia - has not had not enough government budget. Hence, they had to stop those interventionist policies and focus on vaccine distribution and importations. Therefore, it can be concluded that the ability to employ interventionist policies to reduce unemployment depends on the government budget.